RAFFLES INSTITUTION
H2 Computing (9569)

2026 Year 6

Revision Exercise S3C: Structured Query Language

1. A hospital stores information about its patients, staff, and wards using the following tables:

PATIENT *(<u>PatientID</u>, Gender, Name, DateOfBirth, Address, ContactNo)*
STAFF *(<u>StaffID</u>, Name, Gender, DateOfBirth, Address, ContactNo, Grade)*
WARD *(<u>WardNo</u>, Class, NoOfBed, Level, StaffID)*

- PatientID, StaffID, and WardNo are primary keys.
- Each ward is assigned one staff member who is in-charge of the ward.
- All tables have been created successfully unless stated otherwise.

(a) Write an SQL statement to create the STAFF table.
(b) The hospital wants to record the email address of each staff member. Write an SQL statement to add a new field called Email to the STAFF table.
(c) Write an SQL statement to insert the following record into the WARD table:

WardNo	Class	NoOfBed	Level	StaffID
W05	B2	12	3	S013

(d) Write an SQL statement to display the Name and ContactNo of all patients.
(e) Write an SQL statement to display
- the staff member's Name, renamed as StaffName
- the Grade

for all staff members, sorted by Grade in descending order.
(f) Write an SQL statement to update the ContactNo of the patient with PatientID = 'P028' to '91918282'
(g) Write an SQL statement to delete all patients who were born before 1 January 1980.
(h) Write an SQL statement to display
- the WardNo
- the Class
- the Name of the staff member in-charge of the ward

for all wards located on Level 2.
(i) Write an SQL statement to display:
- the Grade
- the total number of staff members in each grade

Show only grades with more than 5 staff members.
(j) Write an SQL statement to display the top 3 wards with the highest number of beds.
(k) Write an SQL statement to remove the WARD table from the database.

2. A hotel manages its bookings, rooms, and customers using the following tables:

CUSTOMER *(<u>CustomerID</u>, Name, Address, Telephone, Email)*
ROOM *(<u>RoomNumber</u>, RoomDescription, RoomType)*
ROOMTYPE *(<u>RoomType</u>, WeekdayRate, WeekendRate)*
BOOKING *(<u>BookingID</u>, CustomerID, RoomNumber, ArrivalDate, DepartureDate)*

- ROOM.RoomType is a foreign key referencing ROOMTYPE.
- BOOKING.CustomerID is a foreign key referencing CUSTOMER.
- BOOKING.RoomNumber is a foreign key referencing ROOM.
- Rates are in dollars per night.
- ArrivalDate and DepartureDate are of DATE type (subtraction gives number of nights)
- All tables have been created successfully unless stated otherwise.

(a) Write an SQL statement to create the ROOMTYPE table with:
- RoomType as the primary key
- WeekdayRate and WeekendRate as numeric fields

(b) The hotel wants to record whether a room has a sea view. Write an SQL statement to add a new column SeaView (YES/NO) to the ROOM table.

(c) Insert the following record into **CUSTOMER**:

CustomerID	Name	Address	Telephone	Email
AB025	Sarah Tan	23 Beach Road	87654321	sarah.tan@mail.com

(d) Write an SQL statement to display:
- Customer Name
- RoomNumber
- ArrivalDate
- DepartureDate

for all bookings arriving in January 2026.

(e) Write an SQL statement to display:
- BookingID
- Customer Name
- TotalAmount = Number of nights × Room rate

Assume the rate comes from ROOMTYPE.WeekdayRate for all nights.

(f) A room type's rate has increased. Write an SQL statement to update all rooms of type 'Deluxe' so that WeekdayRate is now 150.

(g) Write an SQL statement to delete all bookings where DepartureDate is before 1 January 2026.

(h) Write an SQL statement to display:
- Customer Name
- Total amount spent across all bookings as TotalSpent

Show the top 3 customers by amount spent.

(i) Write an SQL statement to display number of bookings made for sea view rooms in December 2025.